



Magnetic Activities of AD Leonis: Flares in TESS Data and Optical Spectra

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Abstract

We studied the flaring activities of M4.5 dwarf AD Leo to understand its stellar atmospheres and magnetic activities. We present new observational results on this highly active star, using nonsimultaneous measurement of Transiting Exoplanet Survey Satellite (TESS) data, time-series optical spectra, and Giant Meter Radio Telescope 325 MHz radio data. We revisited the rotation period of 2.23 ± 0.04 days from TESS, which matches well with previously measured literature values. We estimated an extremely rare high-energy superflare of 4.9×10^{35} erg and ~ 400 minute duration with a high magnetic field strength of 1.2 kG. Interestingly, we correlated the duration of a flare event with its energy, $\tau \propto E_{\text{bol}}^{0.60 \pm 0.02}$, and noticed a discrepancy between stellar and solar flares suggesting a difference in coronal magnetic field strength. From time-series spectra, we observed H α spectral flares in the range of 10^{30} – 10^{31} erg. A 12 minute delay in a spectral flare event was observed between the emission of the H α and Ca II H and K lines, possibly due to their origination at different spatial locations in the chromosphere. We noticed the deviation in flare rate distribution and orbital phases indicates the presence of highly active regions. Furthermore, an occasional radio detection with a flux density of 9.46 ± 1.63 mJy at a frequency of 325 MHz might be coherent emission in the presence of the magnetic field, giving a hint of star–planet interaction.

Unified Astronomy Thesaurus concepts: M dwarf stars (982); Stellar flares (1603); Magnetic fields (994); Starspots (1572); Habitable planets (695)

1. Introduction

The main-sequence M dwarfs are interesting targets in the search for habitable planets and life beyond our solar system (J. Scalo et al. 2007). Over 5000 confirmed exoplanets have been discovered until now. Many of these planets are placed in their star’s canonical habitable zone, which is the region around a star where an orbiting planet can support an Earth-like atmosphere (M. H. Hart 1979; A. L. Shields et al. 2016a). The interaction between a star and its orbiting planets can induce both radiative and gravitational effects on planetary climate, such as albedo, tidal evolution, melting of ice sheets, etc. (M. I. Budyko 1969; R. Barnes et al. 2008; A. L. Shields et al. 2016b), which enters into the larger discussion of habitability (A. L. Shields et al. 2016a). M dwarfs are the smallest cool hydrogen-burning objects, with masses in the range of 0.075 – $0.60 M_{\odot}$ (I. Baraffe & G. Chabrier 1996) and effective temperatures in the range 2400 – 3800 K, and the most abundant and long-lived objects (G. Laughlin et al. 1997), comprising $\sim 75\%$ of the total stellar population, within the Milky Way (T. J. Henry 2007; J. J. Bochanski et al. 2010). The chances of the occurrence of planetary systems, especially Earth-like planets orbiting in “habitable zones,” increase with decreasing stellar mass and radius, which is confirmed by observational evidence (A. W. Howard et al. 2012).

The magnetic fields in the Sun and Sun-like stars are believed to be generated in their interior through a magneto-hydrodynamic (MHD) dynamo mechanism that involves interactions between plasma flows and magnetic fields (P. Charbonneau 2020). The efficiency of the magnetic field generation in such stars is determined by the interaction

between differential rotation and subphotospheric convection into the stellar interior, where meridional circulation plays an important role (A. S. Brun et al. 2015; P. Charbonneau 2020). A turbulent dynamo might generate the magnetic fields in the fully convective M dwarf stars, which leads to more frequent and powerful flaring activities (G. Chabrier & I. Baraffe 1997; J. R. A. Davenport et al. 2014). In stellar activity, the magnetic fields reach the stellar surface and manifest themselves in a variety of phenomena, i.e., starspots, chromospheric plagues, and heating of the chromosphere and corona with impulsive flares. Impulsive flares are visible in all regions of the spectrum due to the reconnection of magnetic field lines (A. Skumanich et al. 1975; C. J. Schrijver et al. 1989; S. K. Solanki et al. 2006; A. R. Choudhuri 2017). The magnetic activity increases with decreasing stellar mass, and the stellar magnetic field is responsible for the magnetic activity of late-type main-sequence stars (S. L. Hawley et al. 1996; A. A. West et al. 2008; E. R. Newton et al. 2017).

AD Leonis (AD Leo, GJ 388, BD +20 2465, TIC 95431305) is one of the closest active M dwarfs at a distance of only 4.9 pc in the northern hemisphere. It is a well-known flaring star (N. M. Hunt-Walker et al. 2012; A. P. Buccino et al. 2014) with spectral type M4.5V (M. Tuomi et al. 2018). The most important properties of this star are shown in Table 1. The rotation period of AD Leo has been measured accurately several times, and one such accurate rotation period is estimated to be 2.2399 ± 0.0006 days from the ESPaDOnS and NARVAL spectropolarimetric observations (J. Morin et al. 2008) with an inclination of the rotation axis of $15^{+2.5}_{-2.0}$ (M. Tuomi et al. 2018). Similar to the 11 yr magnetic activity cycle of the Sun, AD Leo shows a possible magnetic activity cycle. A. P. Buccino et al. (2014) estimated an activity cycle with a period of 2769 ± 106 days (~ 7.58 yr) based on long time-series Ca II K lines in the CASLEO spectra, and another possible period of 2937 ± 575 days (~ 8.04 yr) based on both long time-series



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Table 1
Stellar Parameters of AD Leo

Parameter	Value	Reference
BD	+20 2465	(1)
GJ	388	(2)
TIC	95431305	(3)
R.A. (epoch J2016.0)	10:19:35.7	(4)
Decl. (epoch J2016.0)	19:52:11.3	(4)
d (pc)	4.964 ± 0.001	(4)
Galactic population	Young disk	(5)
Stellar kinematic group	Castor	(6)
G (mag)	8.2041 ± 0.0015	(4)
J (mag)	5.449 ± 0.027	(7)
V (mag)	9.52	(8)
T (mag)	7.04	(9)
Spectral type	M4.5V	(10)
T_{eff} (K)	3414 ± 100	(11)
$\log g$ (cgs)	5.12 ± 0.12	(12)
[Fe/H] (dex)	0.28 ± 0.17	(13)
$< B >$ (G)	3357 ± 172	(14)
Luminosity	$0.0227 L_{\odot} \pm 0.0003$	(15)
R ($R_{\odot}$)	$0.436 \pm 0.049 R_{\odot}$	(11)
M ($M_{\odot}$)	$0.41 M_{\odot} \pm 0.04$	(15)
$v \sin i$ (km s $^{-1}$)	2.4 ± 1.5	(16)
P_{rot} (day)	$2.23^{+0.36}_{-0.27}$	(17)
i (deg)	$15^{+2.5}_{-2.0}$	(10)

References. (1) F. W. A. Argelander (1861); (2) W. Gliese (1957); (3) K. G. Stassun et al. (2018); (4) A. Marecki (2021); (5) D. Montes et al. (2001); (6) J. López-Santiago et al. (2010); (7) M. F. Skrutskie et al. (2006); (8) N. Zacharias et al. (2013); (9) T. Han & T. D. Brandt (2023); (10) M. Tuomi et al. (2018); (11) E. R. Houdebine et al. (2016); (12) E. Marfil et al. (2021); (13) B. Rojas-Ayala et al. (2012); (14) A. Reiners et al. (2022); (15) A. W. Mann et al. (2015); (16) D. Kossakowski et al. (2022); (17) N. M. Hunt-Walker et al. (2012).

ASAS-S optical photometry and CASLEO spectra. M. Tuomi et al. (2018) estimated also that the long-term activity cycle of AD Leo is 4070 ± 120 days (~ 11.14 yr) based on ASAS-N and ASAS-S V-band photometry data. Therefore, the long-term magnetic activity cycle of AD Leo is estimated to be in the range of 7–11 yr based on previous observations.

Solar and stellar flares are electromagnetic explosions that last from a few seconds to hours in the stellar corona and are observable evidence of magnetic energy released. Magnetic reconnection plays an important role in reconfiguring the magnetic field lines and converting magnetic energy into kinetic and thermal energies of plasma (T. G. Forbes & L. W. Acton 1996). A strong magnetic field is detected in AD Leo in the few hundred gauss to kilogauss range (J. Morin et al. 2008). From a high-resolution infrared spectrum of AD Leo, obtained with the Kitt Peak 4 m Fourier Transform Spectrometer, the maximum detected magnetic field strength is estimated to be 3800 ± 260 G in the active regions covering about 73% of its surface, while the surface average magnetic field strength is 2800 G (S. H. Saar & J. L. Linsky 1985). The frequent strong stellar flares on AD Leo have been observed along the entire electromagnetic spectrum, in the optical wavelength (S. L. Hawley et al. 1995; I. Crespo-Chacón et al. 2006), X-ray wavelength (T. R. Ayres et al. 1979), UV wavelength (D. J. Christian et al. 2006), and radio wavelength (J. Villadsen & G. Hallinan 2019).

A radio burst at 1400 MHz with 100% circular polarization from Very Large Array (VLA) observation and low-frequency

radio emission at 144 MHz from the Low-Frequency Array (LOFAR) was observed on AD Leo (K. R. Lang et al. 1983; J. R. Callingham et al. 2021). Detection of low-frequency (144 MHz) radio emission using LOFAR in an exoplanet host M dwarf, GJ 1151, has been interpreted as evidence of star–planet interaction (SPI; B. J. S. Pope et al. 2021), and a similar interpretation for AD Leo was proposed by J. R. Callingham et al. (2021). M. Tuomi et al. (2018) first proposed the existence of a possible planet around AD Leo with a mass of $0.237 \pm 0.047 M_{\text{Jup}}$ or $75.4 \pm 0.047 M_{\oplus}$, at a distance of 0.143 au with 1:1 spin–orbit resonance based on the High Accuracy Radial Velocity Planet Searcher (HARPS) radial velocity (RV) data. However, I. Carleo et al. (2020), using STELLA and HARPS-N RV data, P. Robertson et al. (2020), using Habitable Zone Planet Finder (HPF) RV data, and A. Carmona et al. (2023), using SPIRou and SOPHIE RV data, disputed the argument proposed by M. Tuomi et al. (2018). D. Kossakowski et al. (2022) argued that it is not possible to determinedly prove that there is no planet orbiting in synchronization with the stellar rotation of AD Leo based on CARMENS, HARPS, HIRES, GIARPS, and HPF RV data.

AD Leo is a highly magnetically active and well-known flaring star, thus we aim here to study the magnetic activities based on its flaring activities. We investigate the flaring activity using a 2 minute short-cadence Transiting Exoplanet Survey Satellite (TESS) photometric light curve, new time-series optical spectroscopic data by the 2 m Himalayan Chandra Telescope (HCT), and low-frequency radio emission using Giant Metrewave Radio Telescope (GMRT) 325 MHz data. This paper is organized as follows. We describe the observations and data reduction in Section 2. In Section 3, we report our obtained results from the studies. In Section 4, we discuss our obtained results from the studies. Finally, we have summarized our work in Section 5.

2. Observations and Data Reduction

2.1. Photometry

The TESS satellite was launched on 2018 April 18, with the primary goal of searching for transiting exoplanets around the brighter stars across the entire sky. The TESS has four telescopes of size 10.5 cm with wide-field cameras in a red (600–1000 nm) bandpass that continuously observes a $24 \times 96 \text{ deg}^2$ strip in the sky for 27 days consecutively (see G. R. Ricker et al. 2014 for details).

AD Leo (TIC 95431305, $T_{\text{mag}} = 7.04$; T. Han & T. D. Brandt 2023) was observed in four of the TESS sectors: in sector 45 (2021 November 6–2021 December 2), sector 46 (2021 December 2–2021 December 30), sector 48 (2022 January 28–2022 February 26), and sector 72 (2023 November 11–2023 December 7), respectively. However, data from TESS sectors 45, 46, and 72 are unsuitable for scientific analysis due to the saturation of the pixels, probably caused by a nearby bright star, and we did not consider them in this work. The other sector, 48, provides valuable scientific data and is considered in this work for further analysis. We have selected a 2 minute cadence of TESS sector 48 data in two orbits (orbits 103 and 104) as a part of its Cycle 4 observations. First, we use the “lightcurve” (Lightcurve Collaboration et al. 2018) package to download the TESS light curve of our target for sector 48 from the Mikulski Archive for Space Telescopes (MAST). Then, the Science Processing Operations Center

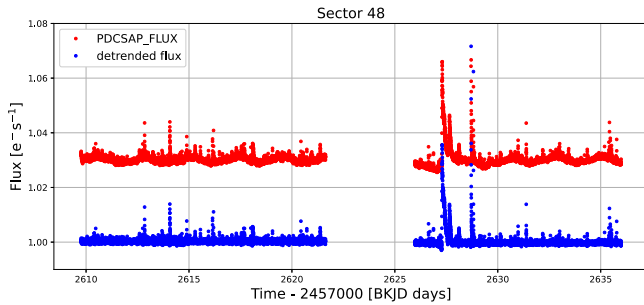


Figure 1. TESS PDCSAP light curve (red) and detrended light curve (blue) of AD Leo in sector 48.

(SPOC) pipeline (J. M. Jenkins et al. 2016) extracts the 2 minute cadence data. We have chosen the presearch data conditioning simple aperture photometry (PDCSAP) light curve in our analysis (J. C. Smith et al. 2012; M. C. Stumpe et al. 2012), which is the simple aperture photometry (SAP) flux after removing the long-term trends using cotrending basis vectors. The “presearch data conditioning” (PDC) module also recognizes and records outliers and makes corrections for the crowding contamination from surrounding nearby stars, where SAP flux does not (N. M. Guerrero et al. 2021). We have taken only the data with $QUALITY = 0$, while a nonzero value of $QUALITY$ indicates that the data have been compromised to some degree by instrumental effects. We also removed the outliers and undefined values. We generate the normalized light curve by dividing each point’s flux by the target mean flux. The PDCSAP flux light curve is shown in Figure 1. The SPOC pipeline corrects for the dilution from other nearby stars in and around the TESS aperture, as denoted by the CROWDSAP value in the TESS header file. The value of CROWDSAP is 0.99 in TESS sector 48, which indicates how much flux is due to the object only in the selected aperture. Therefore, there is no potential effect of nearby bright stars.

2.2. Optical Spectroscopy Data

The optical spectra were taken using the Hanle Faint Object Spectrograph Camera (HFOSC) instrument mounted on the 2 m HCT located at Hanle, India. The HFOSC instrument has several gratings covering different wavelength ranges and resolutions. We have used grism 7 with a long slit having dimensions of $11' (L) \times 1.92' (W)$ for our studies, which covers the wavelength range of 370–720 nm and provides a resolution $R \sim 1200$.³ The observational details are presented in Table 2. The wavelength calibration was carried out using the arc lamp (FeAr lamp), and the flux was calibrated using the standard star Feige 34.

The imaging data reduction was made using the software package Image Reduction and Analysis Facility (IRAF⁴). The preprocessing steps like trimming, bias subtraction, and flat-fielding were performed. The one-dimensional spectrum was extracted using the IRAF task APALL. This APALL task includes defining the aperture, the spectrum extraction, and tracing the spectrum along the dispersion axis. The calibration lamp spectra were extracted regarding the respective target source. The atomic emission lines in the calibration lamp spectra were identified and the pixel-to-wavelength mapping

Table 2

Log of Optical Spectroscopic Observations of AD Leo on 2 m HCT

Date of Observation	No. of Spectra	Exposure Time (s)	Run Time Length (UTC)	Total Observation Time (minutes)
2022-12-02	14	60	20:56–23:59	183
2023-03-11	39	60	14:01–22:41	520

was done using the tasks IDENTIFY and DISPCOR, respectively. The wavelength-calibrated spectra were also flux-calibrated using the standard tasks and calibrated with the standard star spectrum taken on the same night. The wavelength-calibrated and flux-calibrated spectra are shown in Figure 2. We measured the equivalent width (EW) of the spectral lines using the standard equation:

$$EW_{\lambda} = \int_{\lambda_1}^{\lambda_2} \left[1 - \frac{F(\lambda)}{F_c(\lambda)} \right] d\lambda. \quad (1)$$

Here, $F(\lambda)$ represents the flux across the wavelength range of the line ($\lambda_2 - \lambda_1$), and $F_c(\lambda)$ stands for the estimated continuum flux on either side of the absorption line. The EW is measured using the PYTHON-based package SPECUTILS⁵ for our observed object. The signal-to-noise ratio (SNR) of the optical spectra and the errors associated with the flux measurements were derived using a simple spectroscopic SNR measurement algorithm DER SNR.⁶ The features and the continuum points used to calculate the EWs are given in Table 3.

2.3. Radio Observations Using the GMRT

Low-frequency radio continuum observations at 325 MHz were carried out using the GMRT⁷ on 2006 November 24 and 2006 December 3. Further details of the observations are listed in Table 4. We used GMRT archival data (PI: R. A. Osten, Proposal ID: 11ROA01). We used the Astronomical Image Processing System⁸ (AIPS) software package for data reduction. Bad data were flagged, and the remaining data were calibrated and imaged. The AIPS task IMAGR was used to make the final “clean” images.

3. Results

3.1. Period Analysis of the TESS Light Curve

Periodic variation in light curves of low-mass M dwarfs can be observed by rotational modulation of the stellar flux due to starspots, which are cooler than its surroundings and rotating in and out of view (A. McQuillan et al. 2013; L. Doyle et al. 2019). To search for a significant periodic signal, we used the Lomb–Scargle (LS) periodogram package from the NASA Exoplanet Archive Periodogram Service⁹ to find the significant periodic signal and the generalized Lomb–Scargle periodogram¹⁰ (GLS; M. Zechmeister & M. Kürster 2009). The LS periodogram (N. R. Lomb 1976; J. D. Scargle 1982) method uses a Fourier power spectrum estimator to find out significant periodicity, and it can deal with the observed sparse data sets.

⁵ <https://github.com/astrophy/specutils>

⁶ https://www.stecf.org/software/ASTROsoft/DER_SNR/

⁷ <https://www.gmrt.ncra.tifr.res.in/>

⁸ <http://www.aips.nrao.edu/>

⁹ <https://exoplanetarchive.ipac.caltech.edu/>

¹⁰ <https://github.com/mzechmeister/GLS>

³ <https://www.iap.res.in/centers/iao/facilities/hct/hfosc/>

⁴ <https://iraf-community.github.io>

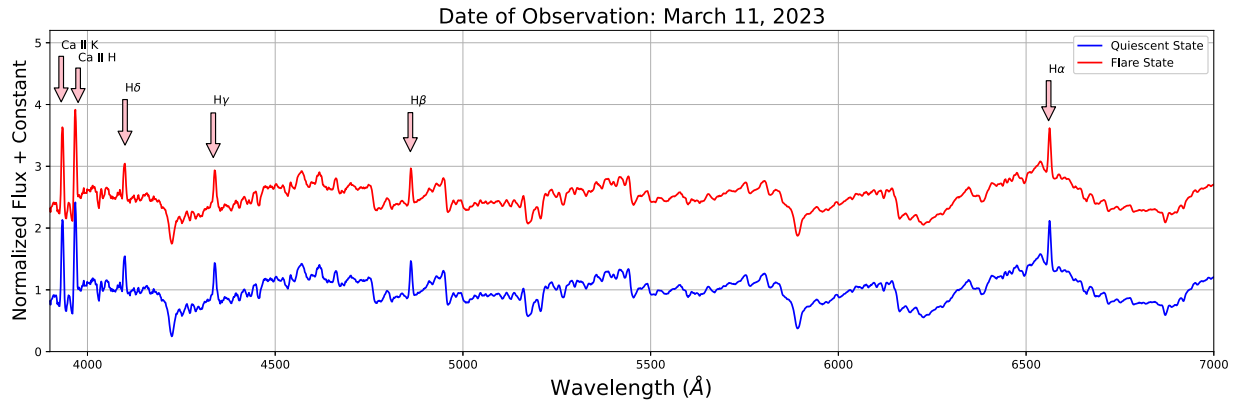


Figure 2. Observed spectrum of AD Leo at the flare state (red) detected in 2 m HCT and its quiescent state (blue) on 2023 March 11. The chromospheric lines are identified.

Table 3
Optical Spectral Features and Continuum Points in Estimated EWs

Feature	Feature Window (μm)	Blue Continuum (μm)	Red Continuum (μm)
Ca II K (0.393 μm)	0.3926–0.3941	0.3918–0.3926	0.3941–0.3950
Ca II H (0.396 μm)	0.3963–0.3977	0.3955–0.3963	0.3977–0.3985
H α (0.656 μm)	0.6552–0.6578	0.6542–0.6552	0.6578–0.6588
H β (0.486 μm)	0.4855–0.4873	0.4845–0.4855	0.4873–0.4883
H γ (0.434 μm)	0.4328–0.4354	0.4315–0.4328	0.4354–0.4363
H δ (0.410 μm)	0.4094–0.4110	0.4085–0.4094	0.4110–0.4118

Note. Wavelengths are given in vacuum.

Table 4
Log of Radio Observations of AD Leo on GMRT

Observation Date	Central Observing Frequency (MHz)	Total Time Spent On- source (minutes)	Flux Calibrator	Phase Calibrator
2006-11-24	325	295	3C 147	1021+219
2006-12-03	325	558	3C 147	1021+219

The rotation period of AD Leo is estimated to be 2.20 days in sector 48 from the NASA Exoplanet Archive Periodogram Service and 2.205 ± 0.002 days using the GLS. We also checked the period using the publicly available STARLINK¹¹ software database, which includes systematic errors. The rotation period of AD Leo is estimated to be 2.23 ± 0.04 days in sector 48 using the Starlink software, and the estimated period by Starlink is adopted here. All the periodograms in three different software programs provide fairly close values of significant peaks in our analyses. The rotation period matches well in all previously measured literature values as mentioned in Table 5, including our new measurements. The LS periodogram and the phase light curve of AD Leo are shown in Figure 3 from the TESS data.

3.2. Detection of Stellar Flare in TESS

First, we have detrended the PDCSAP light curve to remove rotational modulation trends using the Savitzky–Golay filter

Table 5
Rotation Periods of AD Leo

Data	P_{rot} (days)	References
TESS-sector48	2.233 ± 0.047	This work
TESS-sector48	2.2304 ± 0.0401	(1)
ESPaDOnS + NARVAL	2.2399 ± 0.0006	(2)
MOST	$2.23^{+0.36}_{-0.27}$	(3)
ASAS-N	$2.22791^{+0.00066}_{-0.00055}$	(4)
MOST-1	2.289 ± 0.019	(4)
HARPS + HIRES	$2.22567^{+0.00026}_{-0.00011}$	(4)
HP2	2.224	(5)
G1 + G2	2.2246 ± 0.0219	(6)
HT1	2.2244 ± 0.0010	(6)
HT2	2.2225 ± 0.0044	(6)
STELLA	2.237 ± 0.035	(6)
HARPS	2.227	(7)

References. This work: the P_{rot} determined by us from Starlink software; (1) D. Kossakowski et al. (2022); (2) J. Morin et al. (2008); (3) N. M. Hunt-Walker et al. (2012); (4) M. Tuomi et al. (2018); (5) P. Robertson et al. (2020); (6) I. Carleo et al. (2020); (7) A. Reiners et al. (2013).

(“savgol” package). The “savgol” package is computationally fast and gives reliable results in the TESS light-curve analysis. Then, the stellar flares in the detrended light curve were identified using the task *find_flares()* under the open-source Python software ALTAIPONY¹² package. We have identified potential flares only that consist of two or more consecutive flagged points above 2.5σ higher than the mean (L. Doyle et al. 2018, 2019). We have identified 96 flaring events in the detrended TESS light curve in sector 48. The observed flare events in TESS are shown in Figure 4. To calculate the energy of a flare, we first estimated its equivalent duration (ED). It is defined as the equivalent time that it takes for a star to release the same amount of energy at the quiescent stage as it releases during a flare (R. E. Gershberg 1972). It can be computed as the area under the flare light curve in the unit of time.

$$\text{ED} = \int \frac{F_{\text{flare}}(t) - F_0}{F_0} dt \quad (2)$$

where F_{flare} is the integrated flare flux and F_0 is the median quiescent flux. It depends on the used filter and properties of

¹¹ <http://starlink.eao.hawaii.edu/starlink>

¹² <https://altaipony.readthedocs.io/en/latest/>

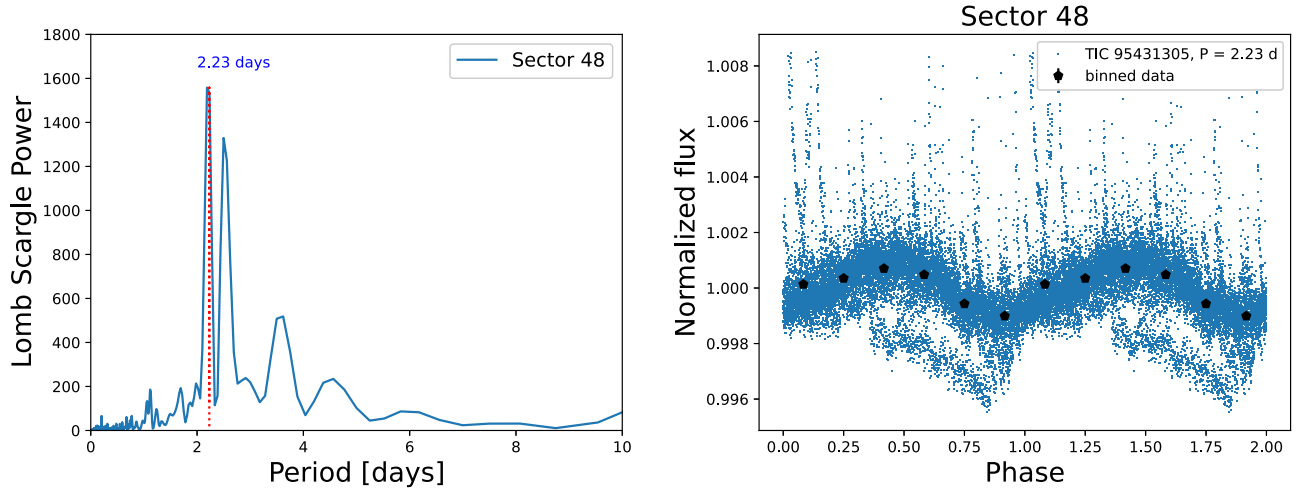


Figure 3. Left panel shows the LS periodogram of AD Leo from TESS (sector 48): the blue line denotes the LS periodogram of AD Leo in TESS sector 48, and the red dashed line denotes the LS periodogram in TESS sector 48 with a period of 2.23 days. Right panel shows a phase folded light curve in sector 48, with the most significant peak obtained from the LS periodogram. Black stars represent the 2500-point binned data.

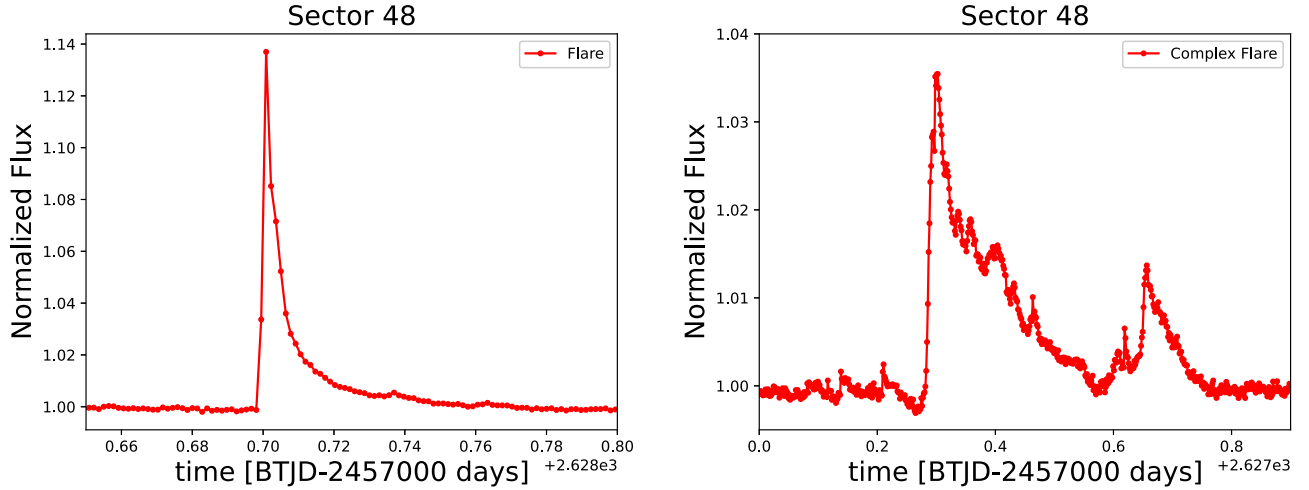


Figure 4. Left panel: normal flare event observed in TESS light curve of AD Leo in sector 48. Right panel: complex flare event of AD Leo in TESS light curve of AD Leo in sector 48.

the photosphere, but not on the distance to the objects (R. R. Paudel et al. 2019). To calculate ED, we used software ALTAIPONY, which automatically detects and characterizes the flares in the object (E. Ilin et al. 2021). Further analysis using ALTAIPONY provides several flare parameters, e.g., start and stop time, flare amplitude and ED, etc. The duration of the flare was calculated using start and stop time. Then, the ED was used to calculate the energy of the flares. To calculate the flare energy E_{flare} , we generated the synthetic photospheric spectrum of our object by using BT-SETTL MODEL.¹³ The fundamental parameters of AD Leo, like the effective temperature (3414 ± 100 K; E. R. Houdebine et al. 2016), surface gravity (5.12 ± 0.12 dex; E. Marfil et al. 2021), and metallicity (0.28 ± 0.17 dex; B. Rojas-Ayala et al. 2012), are used to generate the synthetic spectra. Following S. L. Hawley & G. H. Fisher (1992), we adopted that the observed flare has a temperature T_{flare} of 9000 K. M. Kretzschmar (2011) found that the blackbody spectrum spectral energy distribution of flares can be fitted well with that temperature. Following T. Shibayama et al. (2013), we estimated the bolometric flare luminosity L_{flare}

using the temperature T_{flare} and area A_{flare} of the flare by the Stefan–Boltzmann law,

$$L_{\text{flare}} = \sigma_{\text{SB}} T_{\text{flare}}^4 A_{\text{flare}} \quad (3)$$

where σ_{SB} is the Stefan–Boltzmann constant. The flare amplitude is defined as $\Delta F/F = (F_i - F_o) / F_o$, where F_i is the maximum flux at the flare and F_o is the local mean flux (S. L. Hawley et al. 2014a). Using the flare amplitude (C_{flare}), observed luminosity of the star (L'_{star}), and observed luminosity of the flare (L'_{flare}), we estimate A_{flare} .

$$L'_{\text{star}} = \int R_{\lambda} F_{\lambda}(T_{\text{eff}}) d\lambda \cdot \pi R_{\text{star}}^2 \quad (4)$$

$$L'_{\text{flare}} = \int R_{\lambda} B_{\lambda}(T_{\text{flare}}) d\lambda \cdot A_{\text{flare}} \quad (5)$$

$$C_{\text{flare}} = \frac{L'_{\text{flare}}}{L'_{\text{star}}} = \frac{(F_i - F_o)}{F_o} \quad (6)$$

where λ is the wavelength, $B_{\lambda}(T)$ is the Planck function with each wavelength λ , $F_{\lambda}(T)$ is the flux of the synthetic

¹³ <http://svo2.cab.inta-csic.es/theory/newov2/>

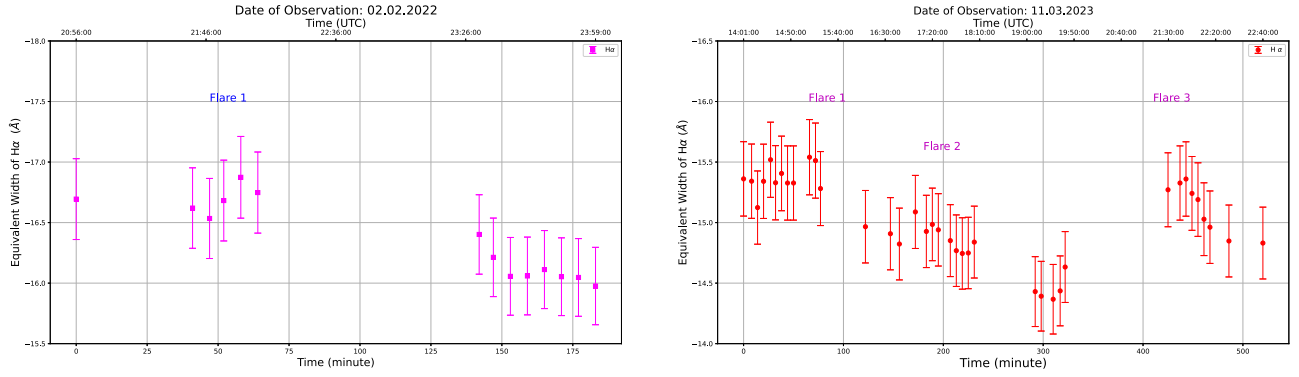


Figure 5. Temporal evolution of the EW of the H α line. The strongest observed flares are labeled.

Table 6
Results of Flare Analysis of AD Leo in TESS in This Work

Sector	Number of Flares	Flares day ⁻¹	Mean Energy (erg)	Lowest Energy (erg)	Highest Energy (erg)
48	96	4.65	1.08×10^{34}	4.86×10^{32}	4.92×10^{35}

photospheric spectrum, and R_λ is the response function of the TESS bandpass. Therefore,

$$A_{\text{flare}} = C_{\text{flare}} \pi R_{\text{star}}^2 \frac{\int R_\lambda F_\lambda(T_{\text{eff}}) d\lambda}{\int R_\lambda B_\lambda(T_{\text{flare}}) d\lambda}. \quad (7)$$

Then the total bolometric energy of the flare (E_{flare}) is an integral of L_{flare} during the flare duration (T. Shibayama et al. 2013):

$$E_{\text{flare}} = \int L_{\text{flare}}(t) dt. \quad (8)$$

Using Equations (3), (7), and (8), we get

$$E_{\text{flare}} = K \cdot (\text{ED}) \quad (9)$$

where K is a constant that depends on the R_λ , B_λ and T_{flare} and ED corresponds to the time-integrated factor of flare amplitude (C_{flare}), which is calculated from the start to end time of the flare (S. L. Hawley et al. 2014a; K. Ikuta et al. 2023).

$$K = \sigma_{\text{SB}} \pi R_{\text{star}}^2 T_{\text{flare}}^4 \frac{\int R_\lambda F_\lambda(T_{\text{eff}}) d\lambda}{\int R_\lambda B_\lambda(T_{\text{flare}}) d\lambda}. \quad (10)$$

We estimated the flare energy using Equation (9), which is in the range of 10^{32} – 10^{35} erg in the identified 96 flare events of TESS sector 48 (see Table 6). J.-Y. Bai et al. (2023) estimated 70 flare events of AD Leo from analysis of the TESS sector 48 data considering slightly different selection criteria for flare events, having three consecutive points above 3σ , while we used here two consecutive points having 2.5σ (L. Doyle et al. 2018, 2019). The bolometric flare energies of AD Leo were measured several times previously, and a few energetic superflares that were reported to be on the order of 10^{34} erg lasted for more than 4 hr on 1985 April 12 (S. L. Hawley & B. R. Pettersen 1991) with a duration of 68 minutes (J.-Y. Bai et al. 2023). In this work, we have estimated for the first time an extremely high superflare event of AD Leo with an energy of 4.9×10^{35} erg and a duration of 400 minutes in TESS sector 48 (see Table 8).

Several multippeak flare events have been observed in TESS sector 48 (see Figure 4). Among 96 flare events in sector 48, two complex flare events with a long decay phase are seen. J.-Y. Bai et al. (2023) estimated the oscillation period of 26.5 minutes during the decay phase in one of the complex events in sector 48. Similar oscillations are observed also in other M dwarfs: Proxima Centauri (K. Vida et al. 2019) and KIC 2852961 (C. E. Pugh et al. 2016). Some of the proposed reasons are MHD oscillations (J. A. McLaughlin et al. 2018) and magnetic dripping (V. M. Nakariakov et al. 2010).

3.3. Spectral Flare Events

Time-series optical spectra of AD Leo were observed over two nights using the HFOSC instrument on the 2 m HCT on 2022 December 2 and 2023 March 11; details are mentioned in Section 2.2 and Table 2. Interestingly, we detected one spectral flare event on 2022 December 2 and three on 2023 March 11. The temporal evolutions of the EWs of the H α , H β , H γ , H δ , and Ca II H and K lines are shown in Figures 5 and 6. Adopting the method of I. Crespo-Chacón et al. (2006), we estimated the fluxes at the quiescent and maximum flaring states using the EWs, and we estimated the line fluxes of the spectral lines, which are converted into luminosities. Following the method of C. Di Maio et al. (2020), we estimated the H α spectral flare energy in the range of 10^{30} – 10^{31} erg (see Table 7 and Figure 7). The H α spectral flare energy has been estimated previously several times, and lies within a range of 10^{29} – 10^{32} erg (I. Crespo-Chacón et al. 2006; C. Di Maio et al. 2020; P. Muheki et al. 2020; K. Namekata et al. 2020). A delay of about 12 minutes was observed between the flare event at the Ca II H and K and the H α lines (see Figure 8). Such delays of 5 ± 3 minutes were noticed earlier (I. Crespo-Chacón et al. 2006; C. Di Maio et al. 2020). Ca II H and K lines are formed at deeper and cooler layers of chromospheric regions than the Balmer series based on the line formation models in stellar atmospheres (I. Crespo-Chacón et al. 2006). Therefore, a possible reason might be the physical origins of those flare events at different chromospheric regions like the Sun (S. S. Hasan & A. A. van Ballegooijen 2008; L. M. Walkowicz & S. L. Hawley 2009).

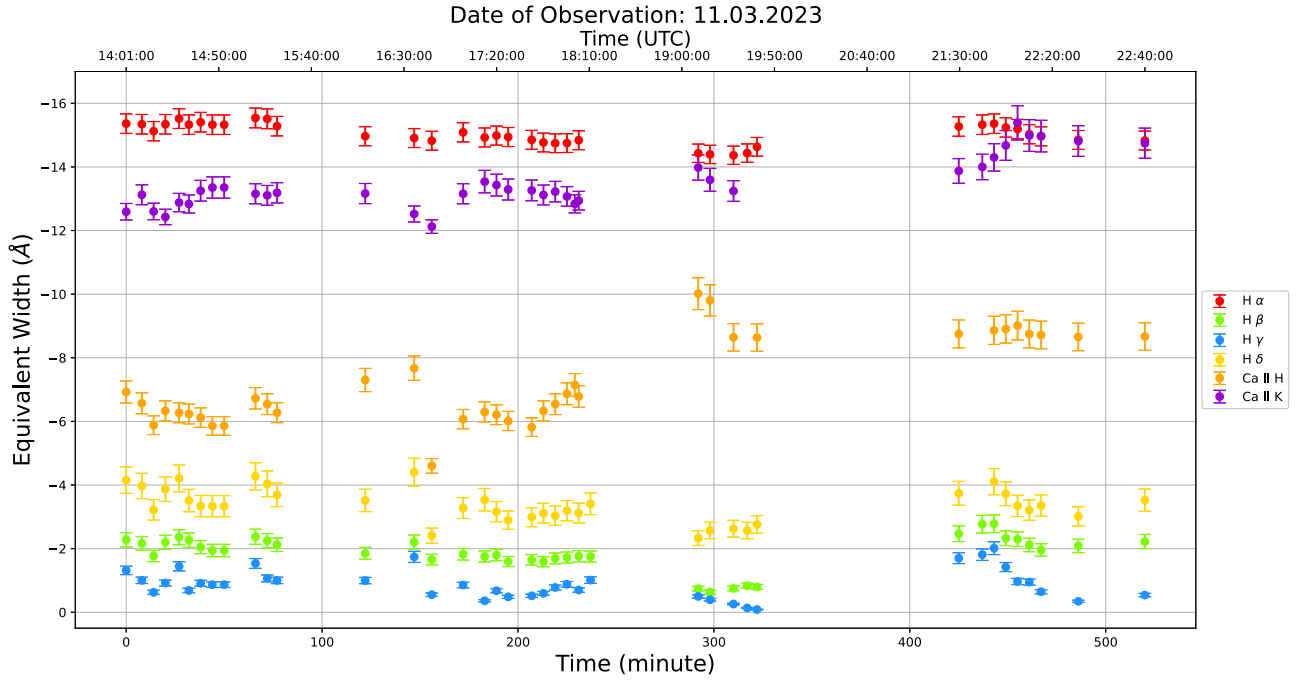


Figure 6. Temporal evolution of the EW of the $H\alpha$, $H\beta$, $H\gamma$, $H\delta$, Ca II H , and Ca II K line.

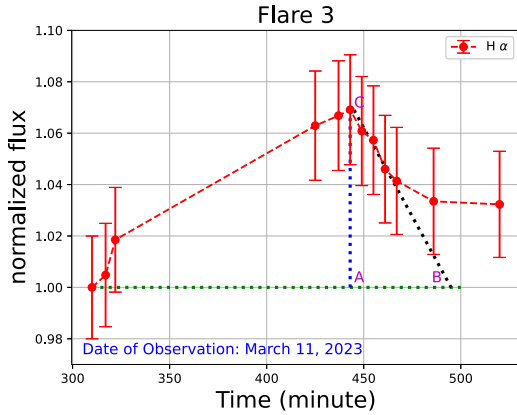


Figure 7. Example of the triangle used to calculate the energy released during the flare in a given line. The blue line, parallel to the ordinate axis and passing through the point of flare maximum, represents the rising phase of the flare; the black line, passing through the two points corresponding to the two observations performed 6 minutes apart, approximates the decay phase of the flare. The green line shows the quiescent state of the star.

3.4. Estimation of Magnetic Field from the Flare Events

Considering similarities between the physical processes involved in the solar flares and stellar flares, several authors obtained the lower limit of the maximum field strength $B_z^{\max}$ using scaling relations between the flare energy and magnetic field (G. Aulanier et al. 2013; R. R. Paudel et al. 2018; Y. Notsu et al. 2019). To estimate the magnetic field from the solar flares, G. Aulanier et al. (2013) assumed that a certain fraction ($\sim 20\%$) of the total available magnetic energy is released by magnetic reconnection through an active region, and the active region is assumed to be associated with positive and negative polarity (bipolar spot) with a linear separation of L^{bipole} . Following the relation given by G. Aulanier et al. (2013) and R. R. Paudel et al. (2018), we can estimate the lower limit of the magnetic field strength ($B_z^{\max}$) to produce

Table 7
Results of the Spectral Flare of AD Leo in This Work

Date of Observation	Flare	Duration (minutes)	$H\alpha$ Flare Energy (erg)
2022.02.02	1	22	3.37×10^{30}
2023.03.11	1	44	1.38×10^{31}
2023.03.11	2	47	5.08×10^{30}
2023.03.11	3	52	2.32×10^{31}

such superflares,

$$E_{\text{flare}} = 0.5 \times 10^{32} \left(\frac{B_z^{\max}}{1000 \text{ G}} \right)^2 \left(\frac{L^{\text{bipole}}}{50 \text{ Mm}} \right)^3 \text{ erg} \quad (11)$$

where E_{flare} is the bolometric flare energy, and L^{bipole} is the linear separation between bipoles. We consider the upper limit of the bipole length, L^{bipole} , equal to πR , as adopted by previous authors (G. Aulanier et al. 2013; R. R. Paudel et al. 2018). R , the radius of AD Leo, is taken to be $0.436 \pm 0.049 R_{\odot}$ (E. R. Houdebine et al. 2016). We have detected 96 flares in sector 48 having flaring energy in the range of 10^{32} – 10^{35} erg, which falls in the superflare category (T. Shibayama et al. 2013). We estimated that the lower limit of the magnetic field strength ($B_z^{\max}$) from Equation (11) for such flare energies is in the range of 40–1200 G in sector 48 (see Table 8).

Following H. Maehara et al. (2015, 2021), we correlate the duration of a flare event (τ) with the bolometric flare energy (E_{bol}) in the left panel of Figure 9. A linear positive correlation is found between τ and E_{bol} for AD Leo, i.e., $\tau \propto E_{\text{bol}}^{0.60 \pm 0.02}$, and this power-law slope of AD Leo is higher than those flares on G dwarfs observed with Kepler (H. Maehara et al. 2015). The right panel of Figure 9 compares τ – E_{bol} relations to solar and stellar flares on various types of stars, and we notice the discrepancy between the duration of flares in AD Leo and solar flares. The discrepancy between the τ – E_{bol} relations was also

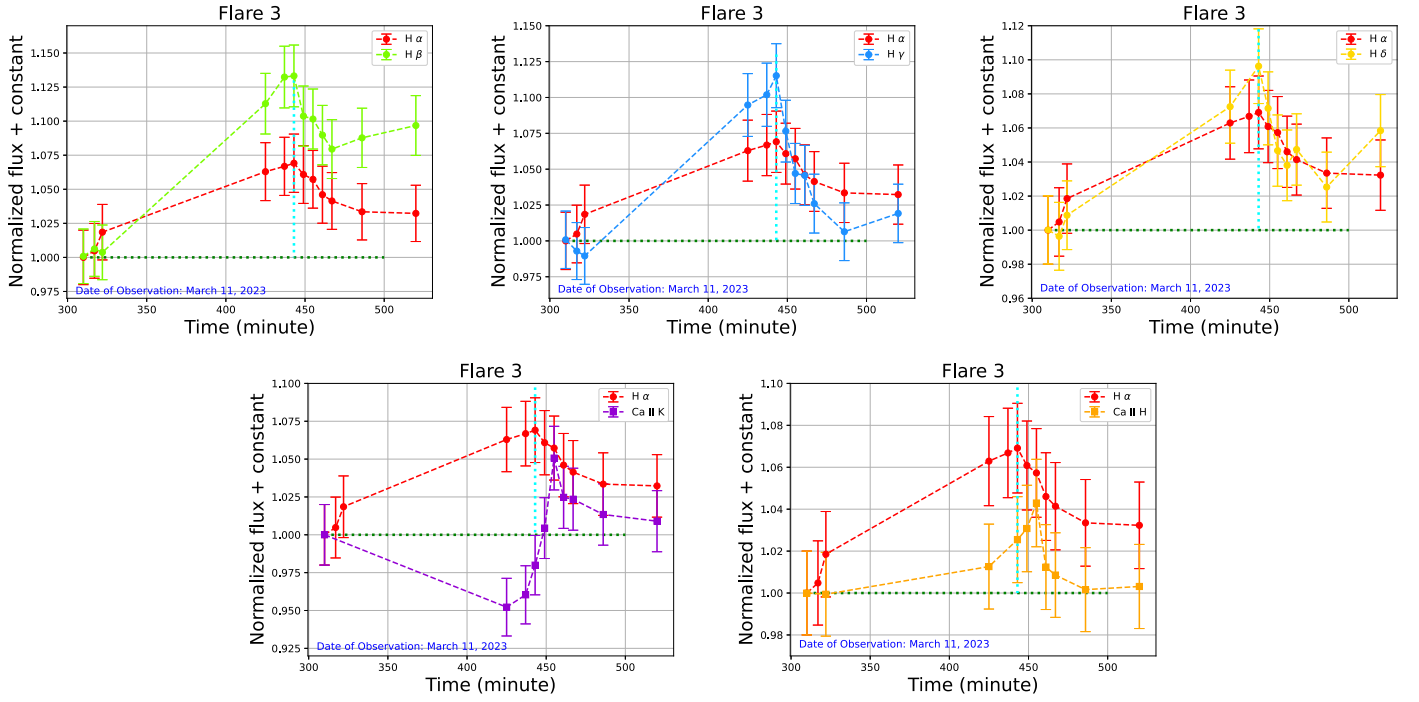


Figure 8. Time series of the normalized flux of analyzed activity indicators to evidence the flare. A delay of 12 minutes is shown in the flare event between the H α and the Ca II H and K lines.

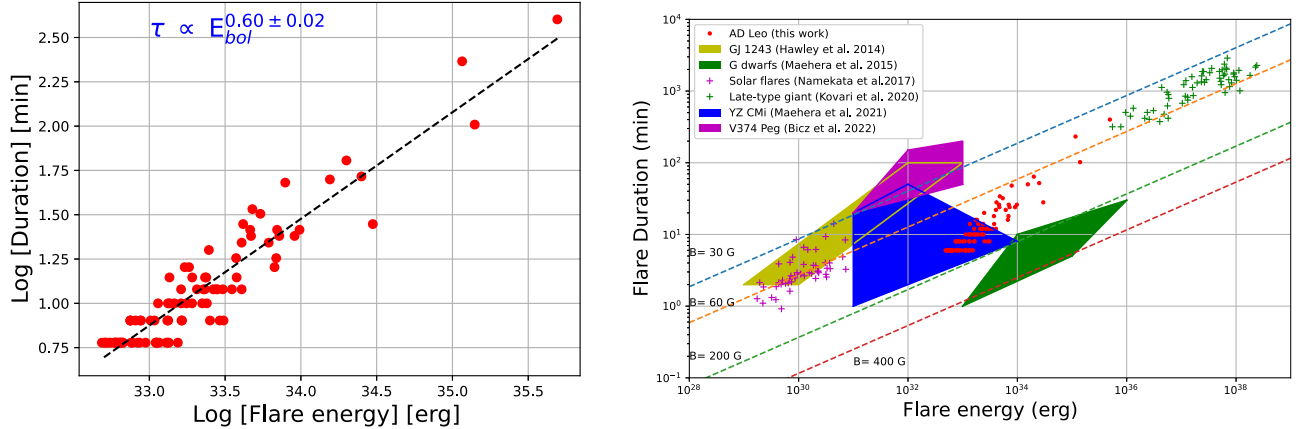


Figure 9. Left panel: scatter plot of flare duration (vertical axis) as a function of the bolometric flare energy (horizontal axis). Right panel: comparison of flare duration and bolometric flare energy (τ - E_{bol}) relations for the stellar flares on M dwarf AD Leo (red filled circles, this work), solar white-light flares (pink crosses; K. Namekata et al. 2017), superflares on solar-type stars (green area; H. Maehara et al. 2015), superflares on late-type giant KIC 2852961 (green crosses; Z. Kővári et al. 2020), and stellar flares on other M dwarfs GJ 1243 (yellow area; S. L. Hawley et al. 2014a), YZ CMi (blue area; H. Maehara et al. 2021), and V374 Peg (dark pink area; K. Bicz et al. 2022). Dashed lines indicate the τ - E_{bol} relation for $B = 30, 60, 200$, and 400 G.

Table 8

Details of the Highest Flare Event of AD Leo in This Work

Observation	Date of Observation	Duration (minutes)	Bolometric Flare Energy (erg)	Magnetic Field (kG)
TESS sector 48	2022.02.16	400	4.9×10^{35}	1.2

found for other M dwarfs, solar white-light flares, and superflares on G dwarfs (S. W. Chang et al. 2015; K. Namekata et al. 2017; W. S. Howard et al. 2019). Therefore, the coronal magnetic field strength difference is one possible explanation

for the discrepancy of the τ - E_{bol} relations in AD Leo, similar to YZ CMi (H. Maehara et al. 2021).

3.5. Area of Starspot

The area of starspots could be determined in several ways, e.g., Zeeman Doppler imaging (L. Rosén et al. 2015), spectral modeling (X.-S. Fang et al. 2016; M. A. Gully-Santiago et al. 2017), planet-transit spot modeling (B. M. Morris et al. 2017), etc. The size of the spot could be estimated from the amplitude and rotational modulation of the light curves (L. M. Rebull et al. 2016; H. A. C. Giles et al. 2017). Following the method described by Y. Notsu et al. (2013), we can estimate the area of the spot using

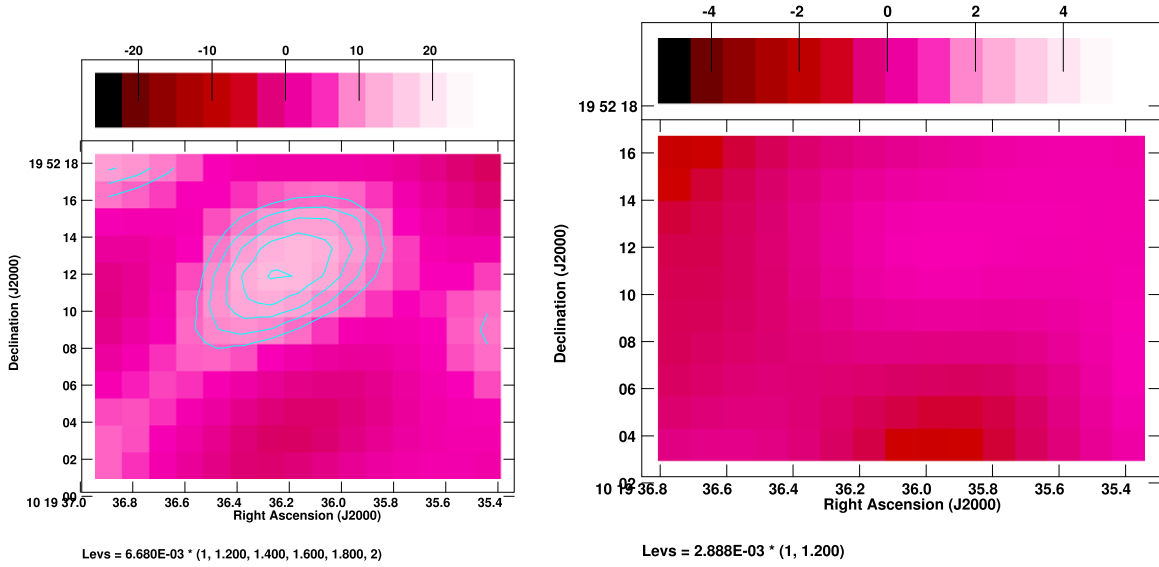


Figure 10. Radio image of AD Leo on 2006 November 24 (left figure) and 2006 December 3 (right figure) at an observing frequency of 325 MHz from the legacy GMRT. The source was detected on 2006 November 24 but not on 2006 December 3. The contour is plotted for the detected case with 5σ level where σ is $1.32 \text{ mJy beam}^{-1}$. For the nondetection case, the contour is plotted with 3σ where σ is $0.96 \text{ mJy beam}^{-1}$.

the following relations,

$$T_{\text{star}} - T_{\text{spot}} = 3.58 \times 10^{-5} T_{\text{star}}^2 + 0.249 T_{\text{star}} - 808 \quad (12)$$

$$A_{\text{spot}} = A_{\text{star}} \left(\frac{\Delta F}{F} \right) \left[1 - \left(\frac{T_{\text{spot}}}{T_{\text{star}}} \right)^4 \right]^{-1} \quad (13)$$

where T_{star} is the effective photospheric temperature and T_{spot} is the spot temperature. The $\Delta F/F$ is the amplitude of the normalized TESS light curve, measured from the phase folded and binned light curve. A_{star} is the area of the stellar disk calculated as $2\pi R^2$ (R is the radius of the star, $0.436 \pm 0.049 R_{\odot}$; E. R. Houdebine et al. 2016). Considering the effective photospheric temperature of AD Leo of 3414 K (E. R. Houdebine et al. 2016), we have estimated T_{spot} to be 2954 K using Equation (12). Using Equation (13), we estimated the approximate spot area to be 1.1% of the total surface area of AD Leo, similar to B. Stelzer et al. (2022).

3.6. Flare Event Rate

AD Leo is a very highly active star, and its flare occurrence rates were estimated several times in the past. In TESS sector 48, we have detected 96 flare events, corresponding to the flare event rate of 0.19 h^{-1} (see Table 6). From new time-series spectroscopic data on 2022 December 2 and 2023 March 11, we estimated the flare event rate of 0.33 h^{-1} and 0.34 h^{-1} , respectively. Previously, the flare event rates were estimated in the range of 0.03 to 0.71 h^{-1} (R. E. Gershberg & P. F. Chugainov 1967; S. Cristaldi et al. 1969; A. H. Jarrett & J. P. Eksteen 1969; S. Cristaldi & M. Rodonò 1970; T. J. Moffett 1974; B. R. Pettersen et al. 1984; R. Konstantinova-Antova & A. P. Antov 1995; I. Crespo-Chacón et al. 2006; N. M. Hunt-Walker et al. 2012; E. W. Guenther et al. 2019; P. Muheki et al. 2020; J.-Y. Bai et al. 2023). We have estimated the time of AD Leo remaining active is 5%–7%, and it was estimated from total flaring and total TESS observation time in sector 48.

3.7. Radio Emission of AD Leo at 325 MHz Using GMRT

We analyzed GMRT archival data at 325 MHz of AD Leo, observed on 2006 November 24 and 2006 December 3 (see Table 4). We estimated a flux density of AD Leo of $9.46 \pm 1.63 \text{ mJy}$ at a frequency of 325 MHz on 2006 November 24 with null-detection on 2006 December 3 (see Figure 10). The 3σ upper limit of the flux density on December 3 is estimated to be 2.88 mJy. The brightness temperature is an important parameter in characterizing the observed radio emission. Following the expression by A. Burrows et al. (2001), the brightness temperature can be expressed as,

$$T_b = 2.5 \times 10^9 \left(\frac{S_{\nu}}{\text{mJy}} \right) \left(\frac{\nu}{\text{GHz}} \right)^{-2} \left(\frac{d}{\text{pc}} \right)^2 \left(\frac{R}{R_{\text{Jup}}} \right)^{-2} \text{ K}. \quad (14)$$

Here, flux density S_{ν} at frequency ν is measured from a source at distance d ($4.964 \pm 0.001 \text{ pc}$; A. Marecki 2021) and emission region of radius R ($0.436 \pm 0.049 R_{\odot}$; E. R. Houdebine et al. 2016). Here, R_{Jup} is the radius of Jupiter, to be $7.0 \times 10^9 \text{ cm}$. The brightness temperature of AD Leo has been estimated several times within the range of 10^{10} – 10^{15} K (A. V. Stepanov et al. 2001; R. A. Osten & T. S. Bastian 2006). We have estimated the brightness temperature to be 10^{11} K on 2006 November 24. The possible reason for this detected radio emission is described in Section 4.3.

4. Discussion

AD Leo is a highly magnetically active and strong flaring object (J. Morin et al. 2008). In this paper, from the TESS light curve and time-series spectroscopic data of AD Leo, we found the bolometric flare energy in a range of 10^{32} – 10^{35} erg (in superflare range) with the lower limit of the magnetic field of 40–1200 G. Such superflare events depend on the size and distribution of the spot area of AD Leo, estimated to be 1.1% of its total surface area, and play an important role in the habitability of any possible nearby planet (V. S. Airapetian et al. 2020; M. Aizawa et al. 2022).

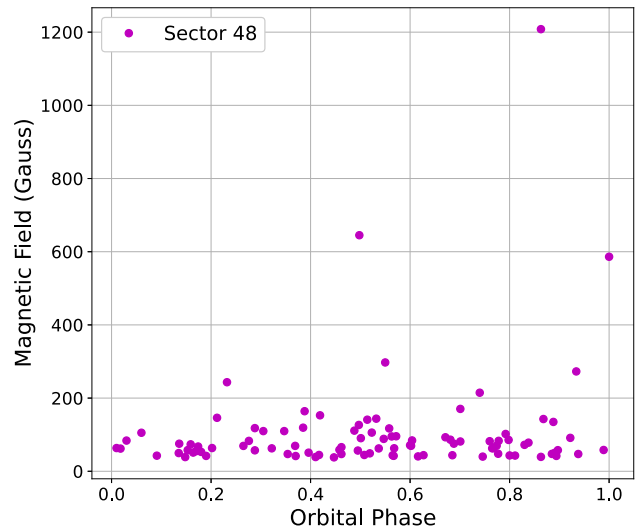
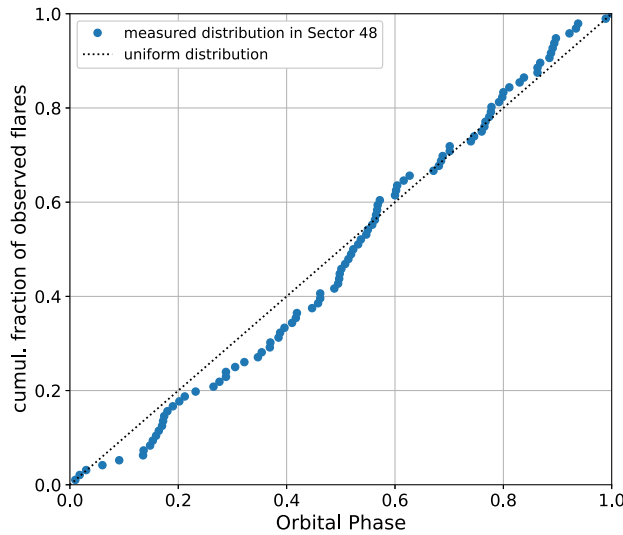


Figure 11. Left panel: cumulative distribution of flares with orbital phase of AD Leo. The underlying distribution is uniform (black dotted line). Right panel: magnetic field of flare events with its orbital phase.

4.1. Correlation of Flare Event Rate with Rotational/Orbital Phases

The variation of flaring rate with a stellar rotational phase can be caused by an extremely active region of the star that rotates in and out of view of the observer. M. Tuomi et al. (2018) proposed that the existence of the Doppler periodicity of AD Leo might be the gravitational tug of a probable planet with a mass of $0.237 \pm 0.047 M_{\text{Jup}}$ or $75.4 \pm 0.047 M_{\oplus}$ orbiting it in 1:1 spin-orbit resonance, and an extreme case of this tidal locking behavior may occur, whereby the rotation period of the star would match the period of planets in their habitable zone (E. R. Newton et al. 2016; A. L. Shields et al. 2016a). We estimated the orbital phases of AD Leo considering spin-orbit resonance by the open-source Python software FLARING-SPI¹⁴ using the “*get_flare_phases()*” function (E. Ilin & K. Poppenhaefer 2022). We estimated the frequency distribution of flares per phase using all valid data points in the detrended light curve of TESS. We observed the deviation from a uniform underlying flare rate distribution with orbital phases (see Figure 11). Flares with lower energies are not completely detected. Therefore, we perform injection and recovery tests to correct the completeness via the detection probability and improve the energy estimation via the energy recovery ratio by ALTAIPONY using the *characterize_flares* function (J. R. A. Davenport 2016; E. Ilin et al. 2021; D.-Y. Gao et al. 2022). There are 96 flare events in TESS sector 48 with a detection probability of 0.88. It is apparent from Figure 11 that most of the flare events’ magnetic fields for sector 48 were found in the range of 40–200 G. However, we noticed some high magnetic fields of superflare events beyond 200 G for sector 48 in the range of orbital phases of 0.23–0.55 and 0.86–1.00 (see Figure 11). The star’s magnetic field might be connected to the magnetic field of the probable planet within the Alfvén surface dissipated via flares (E. Ilin & K. Poppenhaefer 2022). Therefore, it might be a hint of the presence of highly active regions of AD Leo or the SPI with the probable orbiting planet (AD Leo b) around it in this range of orbital phases. However, we cannot rule out the random nature of such high flares.

4.2. Superflares in AD Leo: Habitability of Probable Planet

The flare frequency distribution (FFD) of a star, or the frequency of flares of a given energy, provides information on how much magnetic energy is released through flares. Following the relation of C. H. Lacy et al. (1976), the cumulative FFD can be expressed as

$$\log \nu = \alpha + \beta \log E \quad (15)$$

where ν is the cumulative frequency of the flares with given energy larger than E , α is the constant and β is slope of a linear fit to the cumulative FFD. As the cumulative number of flares per time depends on the ED of the flares, we estimated the value of β by fitting the frequency distribution function with the modified maximum likelihood estimator power law. The best fit gives the value of β as -1.69 ± 0.07 in TESS sector 48. Figure 12 shows the FFDs of AD Leo. The values of slope (β) of AD Leo were measured several times previously, and lie in the range of -0.44 to -1.45 (C. H. Lacy et al. 1976; R. E. Gershberg 1983; B. R. Pettersen et al. 1984; M. Audard et al. 2000; I. Crespo-Chacón et al. 2006; N. M. Hunt-Walker et al. 2012; P. Muheki et al. 2020; J.-Y. Bai et al. 2023). M. N. Günther et al. (2020) estimated the average FFD (slope = -1.4) of 446 M dwarfs based on the TESS data. We noted that the FFD of AD Leo, estimated from the TESS data in this work, is higher than that of previous studies. Following the relation of M. N. Günther et al. (2020), the flare frequency needed to drive prebiotic chemistry or abiogenesis zones (P. B. Rimmer et al. 2018; J. Xu et al. 2018) can be expressed as

$$\nu \geq 25.5 \text{ day}^{-1} \left(\frac{10^{34} \text{ erg}}{E_U} \right) \left(\frac{R}{R_{\odot}} \right)^2 \left(\frac{T_{\text{star}}}{T_{\odot}} \right)^4. \quad (16)$$

Studying the stellar flares from a sample of M dwarfs, M. N. Günther et al. (2020) found that 7.6% of the flare’s bolometric energy falls into the U band, leading to $E_U \approx 7.6\% E_{\text{bol}}$, by using the U -band spectral response function, and the author assumes a 9000 K blackbody for the flare. In recent study, A. F. Kowalski et al. (2025) reported near-ultraviolet flares far from 9000 K blackbody. As AD Leo is a similar spectral type

¹⁴ <https://github.com/ekaterinailin/flaring-spi>

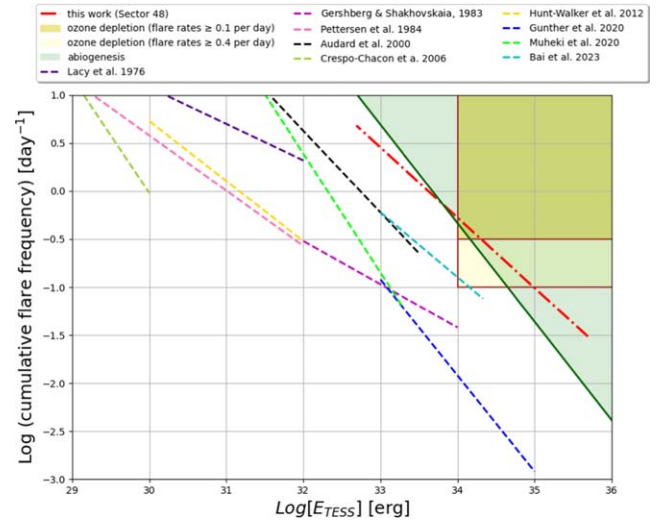
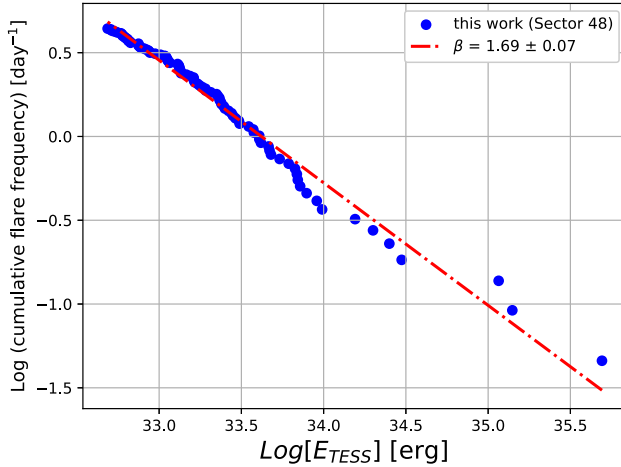


Figure 12. Left panel: cumulative flare frequency distributions (FFDs) obtained from the TESS light curves of AD Leo. Right panel: FFDs of AD Leo compared with the literature values.

object, we can scale the above Equation (16) for our object. We estimate the flare frequency needed to drive the prebiotic chemistry in the probable planet of AD Leo is $7.6 \text{ flares day}^{-1}$ with the bolometric flare energy of $\sim 10^{34} \text{ erg}$ from the above Equation (16). We estimated the flare frequency of $0.34 \text{ flares day}^{-1}$ with the bolometric flare energy of $\sim 10^{34} \text{ erg}$ based on TESS sector 48, which is under the range of the ozone depletion zones (Figure 12). Therefore, the more energetic superflares may also remove the ozone layers, but habitability might exist below an ocean surface on a probable planet of AD Leo (M. A. Tilley et al. 2019).

4.3. Radio Emissions: Probable Star–Planet Interaction

It has long been well known that AD Leo has been very active in radio emissions (S. R. Spangler et al. 1974). Extremely high brightness temperatures ($\geq 10^{15} \text{ K}$) were reported during radio bursts, which suggests that one or more coherent radiation mechanisms, i.e., plasma radiation mechanisms and the electron cyclotron maser instability (ECMI), operate in the corona of AD Leo (R. A. Osten & T. S. Bastian 2006, 2008). High circular polarization (50%–100%) and high brightness temperature ($\sim 10^{11} \text{ K}$) were reported from previous radio observations using VLA and the Five-hundred-meter Aperture Spherical Telescope by the ECMI mechanism (J. Villadsen & G. Hallinan 2019; J. Zhang et al. 2023; X. Zhang et al. 2024). A. Mohan et al. (2024) reported the radio emission (550–850 MHz) using an upgraded Giant Metrewave Radio Telescope with high brightness temperature (10^{10} – 10^{11} K) and 100% left circular polarization by ECMI mechanism. The ECMI radiation indicates sub-Alfvénic interaction between a star and its exoplanet (P. Zarka 2007). Low-frequency (144 MHz) radio emission using LOFAR is also detected in AD Leo, and SPI with its probable orbiting planet was proposed by the authors (J. R. Callingham et al. 2021; J. Zhang et al. 2023). Therefore, our detection of radio emission of AD Leo at 325 MHz from GMRT data with a flux level of $9.46 \pm 1.63 \text{ mJy}$ and brightness temperature of $2.9 \times 10^{11} \text{ K}$ might be due to the ECMI mechanism, and this is possible evidence for SPI.

5. Summary and Conclusion

We present here the results from the light-curve analysis and flaring activities on AD Leo using TESS 2 minute short-cadence photometric data in sector 48, spectral flares using new time-series optical spectra obtained using the 2 m HCT, and low-frequency emission using the GMRT radio data at 325 MHz.

1. The rotation period of AD Leo is estimated to be $2.23 \pm 0.04 \text{ days}$ using the new TESS data, which matches well with the previous estimations. From this short-cadence TESS data, we found 96 flare events in sector 48. Two flare events show multiple or complex peaks, which might be due to quasiperiodic pulsations of the emitting plasma in the chromosphere.
2. We estimated the bolometric flare energy in the range of 10^{32} – 10^{35} erg (in the superflare range). Using the bolometric flare energy based on the TESS flare events, we estimated the lower limit of the magnetic field in the range of 40–1200 G. Such superflare activity depends on the size and distribution of the area of starspots, which is estimated to be 1.1% of the total surface area of AD Leo. We also correlated the duration of a flare event with its energy, $\tau \propto E_{\text{bol}}^{0.60 \pm 0.02}$, and noticed a discrepancy between stellar and solar flares suggesting a difference in coronal magnetic field strength.
3. We estimated the $H\alpha$ flare energy in a range of 10^{30} – 10^{31} erg . A 12 minute delay was observed in a spectral flare event between the Ca II H and K and the $H\alpha$ lines, which might be due to their origination at different spatial locations of the chromosphere.
4. The deviation of a uniform underlying flare rate distribution from TESS with orbital/rotational phases of AD Leo, suggesting a planet in a 1:1 spin–orbit resonance, indicates the presence of highly active regions or might be SPI.
5. Our detection of radio emission of AD Leo at 325 MHz from GMRT data with a flux level of $9.46 \pm 1.63 \text{ mJy}$ and brightness temperature of $2.9 \times 10^{11} \text{ K}$ might be due to ECMI, and this is probable evidence of SPI.

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Data Availability

All the TESS data used in this paper can be found in MAST, DOI:[10.17909/s236-ac55](https://doi.org/10.17909/s236-ac55). The HCT spectroscopic monitoring data may be made available on reasonable request. The corresponding authors may be contacted for that. All the GMRT data used in this paper can be found in the GMRT online archive: <https://naps.ncra.tifr.res.in/goa/data/search>.

Facilities: TESS, HCT, and GMRT.

Software: astropy (Astropy Collaboration et al. 2013, 2018), astroquery (A. Ginsburg et al. 2019), lightkurve (Lightkurve Collaboration et al. 2018), AltaiPony (J. R. A. Davenport 2016; E. Ilin et al. 2021), numpy (C. R. Harris et al. 2020), matplotlib (J. D. Hunter 2007), IRAF (D. Tody 1986), AIPS (E. W. Greisen 1990).

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